

Bank Marketing Analytics



Predicting Customer Subscription to Term Deposit Using Machine Learning....

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ABSTRACT

Understanding customer behavior patterns and predicting whether customers will subscribe to term deposit is really important as it helps to improve conversion rates in bank marketing campaigns and identify the business strategies. The dataset used in this study includes approximately 11 000 observations. Explanatory data analysis and machine learning techniques were used for predicting customers who are really potential to subscription. The dataset includes customer demographic information, customer financial information and details of previous marketing campaigns as predictor variables.

One of the main challenges in improving model performance is data leakage, which has been observed in previous studies reporting inflated accuracy and ROC AUC scores. To mitigate this issue, careful preprocessing was conducted, including the exclusion of the feature that leads to data leakage. Four different models were trained and evaluated: Logistic Regression, Decision Tree, Random Forest, and XGBoost model. The Random Forest model was identified as the best model with 0.75 accuracy and 0.82 ROC AUC score.

Explanatory data analysis and feature importance analysis revealed that customer contact history, outcome of previous campaign and specific demographic factors had a relationship with subscription.

This study was mainly focused on building a realistic and practically applicable model for gaining business insights to improve bank marketing efficiency.

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1. INTRODUCTION

The competitiveness of the bank sector is increasing day by day with the improvement of technology. Especially with the development of data science and data analytics, they are approaching higher achievements than before. Bank marketing is one of the major area where data science has contributed much. In the bank marketing field, direct marketing is used to promote financial products mostly. A term deposit is one of the major financial products that are offered by banks, where customers invest money for a fixed time period in return for a guaranteed interest.

The dataset used in this study includes details about term deposit subscriptions. The data was gathered by the Portuguese Institute. The campaign was conducted through phone calls. The dataset includes a variety of features related to customer demographics, financial status, and previous campaign interactions. The main objective of this study is to build a suitable machine learning model to predict the customer behavior for subscription to term deposits. To achieve this goal, the study aims to perform exploratory data analysis for understanding key factors influencing subscription decisions and understanding customer behaviors. And also build several models like Logistic Regression, Decision Tree, Random Forest, and XGBoost model. And identify the model that is most efficient for this study. Additionally, the study focuses on selecting the most suitable features to improve model performance and interpretability. The study will also focus on providing business insights for improving bank marketing efficiency.

The study helps to design target marketing strategies and supports data-driven decision making. Identifying customers who are more likely to subscribe helps to improve marketing efficiency, reduce operational costs, and increase conversion rates. Furthermore, this study highlights how machine learning can be applied to solve real world problems, especially in understanding customer behavior and improving targeted marketing.

In addition, this study considers important practical challenges like the need for interpretable models. Addressing this ensures that the developed model is not only accurate but also applicable in real world scenarios. Overall, this research contributes to improving the efficiency and effectiveness of bank marketing campaigns through the use of data analytics and machine learning.

2. LITERATURE REVIEW

With the development of machine learning techniques, predictive analytics has contributed to bank marketing to understand customer behavior and improve marketing efficiency. In recent studies, researchers have examined various algorithms, preprocessing techniques, and performance matrices to identify the most effective models for customer subscription to term deposits.

Zaki A.M et al.(2024)[1] examined the role of predictive and machine learning in improving direct marketing strategies for bank term deposits. The study was done using a Kaggle dataset from Portuguese. Several machine learning models were built, including SGD Classifier, K- nearest Neighbours, Logistic regression, Gaussian Naïve Bayes, Decision Tree, and Random Forest. The Random Forest model achieved the best performance with an 87.5% accuracy. The study highlighted how predictive analytics can support banks in enhancing customer targeting and optimizing bank campaigns. The key limitation of this study is restricting real world applicability. The study was mainly targeted at prediction accuracy rather than real world implementation challenges and customer behavior factors.

Similarly, Yi (2024)[2] examined the models for predicting customer subscription to term deposits through Logistic Regression, XGBoost, Neural Networks, and Random Forest. For the study, data cleaning, validation, and forward selection were used, and the findings revealed that Random Forest is the best model with 0.857 ROC AUC value, and the accuracy scores of the models were not mentioned. Same as Ahmed's study, this study also did not focus on understanding customer behaviors and gaining valuable business insights for the banks. In addition, some models have an overfitting risk.

Another study available on Kaggle by Mumford(2023)[3] focused on customer conversion analysis for marketing campaigns using explanatory data analysis. The study revealed that understanding customer demographics, campaign interactions, and behavioral patterns improve marketing conversion rates. Similarly, Adel Gabr(2023)[4] conducted a bank marketing campaign analysis using machine learning approaches to predict customer term deposit subscriptions and achieved 0.84 accuracy for XGBClassifier however, the model is considered less realistic due to the inclusion of variables such as 'duration'. That introduces data leakage and may inflate predictive performance.

3. THEORY AND METHODOLOGY

The main purpose of analyzing this dataset is to build a predictive model for identifying the groups to target and gain some business insights. To succeed the project, several steps were followed.

3.1 Explanatory Data Analysis

The explanatory data analysis was conducted to identify the patterns and trends in the dataset.

3.2 Model Building

Four models were built.

3.2.1 Logistic Regression

Logistic regression is a supervised machine learning algorithm used for binary classification problems. It estimates the probability of an outcome using a linear combination of input features, which is then transformed using the sigmoid function to produce values between 0 and 1. The model is mathematically represented as:

$$P(y = 1) = \frac{1}{1 + e^{-(w^T x + b)}}$$

Where x is input features, w is model weights and b is bias term.

Logistic regression is simple, interpretable, and computationally efficient.

3.2.2 Decision Tree

A decision tree is a supervised learning algorithm that makes predictions by recursively splitting the dataset based on feature conditions. Each internal node represents a decision rule, while each leaf node represents a final prediction. The splitting process continues until a stopping is reached, such as maximum depth or minimum samples per leaf.

Decision Trees are easy to interpret and can handle both numerical and categorical data. It can be prone to overfitting if not properly constrained.

3.2.3 Random Forest

Random forest is an ensemble learning method that builds multiple Decision Trees using different subsets of the training data and features. The final prediction is obtained by aggregating the outputs of all individual trees, typically through majority voting. This approach reduces overfitting and improve generalization performance compare to single Decision tree.

Random forest models are highly accurate and robust but are less interpretable and require more computational resources.

3.2.4 XGBoost

XGBoost is a powerful machine learning algorithm based on gradient boosting decision trees. It builds multiple decision trees sequentially, where each new tree corrects the errors of the previous ones. This model is widely used for classification and prediction tasks. It provides high accuracy, handles missing values efficiently, and reduces overfitting through regularization. It works well with larger datasets.

3.3 Improving Model Performance

3.3.1 Hyperparameter tuning

Hyperparameter tuning was used to select the best parameters for models and improve the performance of the model.

The purpose of the hyperparameter tuning is to improve model performance, to reduce overfitting and underfitting and to find the optimal model configuration for better generalization on unseen data. GridSearch and RandomizedSearch were used.

- **Grid Search:** This is an exhaustive search technique where all possible combinations of predefined hyperparameters are evaluated. Each combination is tested using cross validation and the best performing set is selected based on a chosen evaluation metric. This method was used for logistic regression and decision tree model.
- **Randomized Search.** This method is used for random forest model. This evaluates a fixed number of random combinations from hyperparameter space. Unlike Grid search, it does not test all combinations, making it faster and more efficient for large parameter spaces.

3.3.2 Cross validation in tuning

During hyperparameter tuning, cross validation was used to ensure the model is evaluated on different subsets of the data. This helps in reducing overfitting and provides a more reliable estimate of model performance.

3.3.3 Adjusting threshold

A threshold is the cutoff value used to convert predicted probabilities into class predictions. The default threshold value in binary classification is 0.5. In this study, a lower threshold (0.4) was selected to improve the accuracy of the model.

3.4 Validation

Validation of the study was done through accuracy, precision, recall, F1 score and ROC AUC values.

- **Accuracy:** measures the proportion of correctly classified observations out of all observations.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Where:

TP: True Positives

TN: True Negatives

FP: False Positives

FN: False Negatives

- **Precision:** measures number of customers predicted as deposit subscribers who were actually subscribers.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

- **Recall:** measures the ability of the model to correctly identify actual deposits subscribers.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

- **F1 score :** measures harmonic mean of precision and recall.

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

- **ROC AUC :** measures the ability of the model to distinguish between classes across different threshold values.

4. Description of Data

Table 4.1: Description of the Variables

Variable	type	Description
Age	integer	Age of the customer
Job	categorical	type of job
Marital	categorical	Marital status(whether single, married, divorced)
Education	categorical	Educational level (basic: 4years, basic: 6years, basic: 9years, high school, illiterate, professional course, university degree, unknown)
Default	Binary	Has credit in default?
Balance	Integer	Average yearly balance in euros
Housing	Binary	Have a housing loan?
loan	Binary	Have a personal loan?
Contact	category	Contact communication type(cellular, telephone, unknown)
Month	date	Last contact month of year
Duration	integer	Last contact duration, in seconds
Campaign	integer	Number of contacts performed during this campaign
Pdays	integer	Number of days that passed by after the client was last contacted from a previous campaign
Previous	integer	Number of contacts performed before this campaign
poutcome	category	Outcome of the previous marketing campaign
deposit	binary	Has the client subscribed to a term deposit?

4.1 Data Preprocessing

Missing values and duplicate values were checked and none of them were recorded. Then, the data types were checked.

4.2 Feature Engineering

Under this step, some new variables were created to improve model efficiency and obtain more insights.

- The 'age_group' variable was created by categorizing 'age' variable into five categories as 'student', 'early_carrer', 'mid_carrer', 'pre_retirement', 'retired'.

- The 'balance' variable is skewed, so a log transformation was used for that variable.
- 'Month_sin' and 'month_cos' were created to transform date features into sine and cosine coordinates. The method is the cyclical encoding and ensures the model recognizes that periods like December and January are chronologically close, rather than treating the values as distant numerical extremes.

Then, the encoding methods were used. For this analysis, Ordinal encoding, Target encoding, One-hot encoding, and Boolean encoding were used.

- **Ordinal encoding**

Ordinal encoding is used when categorical variables have a natural order or ranking. In this method, categories are replaced with the integers that reflect their relative order. This can apply to ordinal data and it allows the model to interpret meaningful progression. This method was used for encoding 'education' and 'age_group' variables.

- **Target encoding**

Target encoding is a technique where a categorical value is replaced with the average value of the target variable for the category. This captures the relationship with the target, works well for high cardinality features, reduces the dimension and improves model performance. In this analysis, target encoding was used for 'job' and 'outcome' variables.

- **One hot encoding**

This method is used for nominal categorical variables. Each category is converted into a separate binary feature. This approach ensures that the model does not assume any ranking between categories. Each category is treated independently. However, one-hot encoding can increase dataset dimensionality when variables contain many unique categories. Used this method for 'marital' and 'contact' variables.

- **Boolean encoding**

Boolean encoding converts binary categorical variables into 0 and 1 values to make them usable for machine learning models. Used Boolean encoding for 'default', 'housing', and 'loan'.

The **duration variable** was excluded from the predictive models despite its high predictive power. This variable represents the length of the last contact call and is only available after customer interaction has occurred. Including duration variable may lead to unrealistic model performance and data leakage in real world deployment scenarios. Therefore, the variable was removed to develop a more practical and generalizable prediction model.

Also dropped 'month', 'day', and 'age' variables. Then features and the target variables are specified. The dataset was split into train and test data. Using standardscaler, standardization was done.

StandardScaler : This method helps when features are on different scales, models perform better when data is standardized. Prevents large value features from being dominant. StandardScaler transforms data so that all features have a mean 0 and standard deviation 1, making them comparable for machine learning models.

5. Explanatory Data Analysis

The dataset includes 11162 observations and 17 different variables. The data was mainly gathered across demographic information of the customer, financial information of the customer, and campaign details. The target variable is the ‘deposit,’ which includes binary answers.

5.1 Univariate Analysis

Table 4.1: Summary Statistics of the quantitative variables

	Age	Balance	Duration	Campaign
Count	11162	11162	11162	11162
Mean	41	1528.54	371.99	2.5
Std	11.91	3225.41	347.13	2.72
Min	18	-6847.00	2.0	1.00
25%	32	122.00	138.00	1.00
50%	39	550.00	255.00	2.00
75%	49	1708.00	496.00	3.00
max	95	81204.00	3881.00	63.00

Figure 5.1 : distribution of the subscription (target variable)

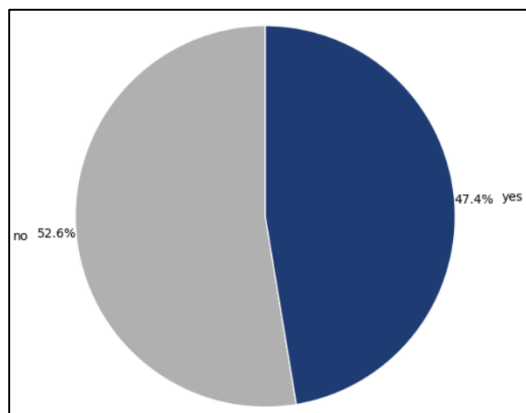
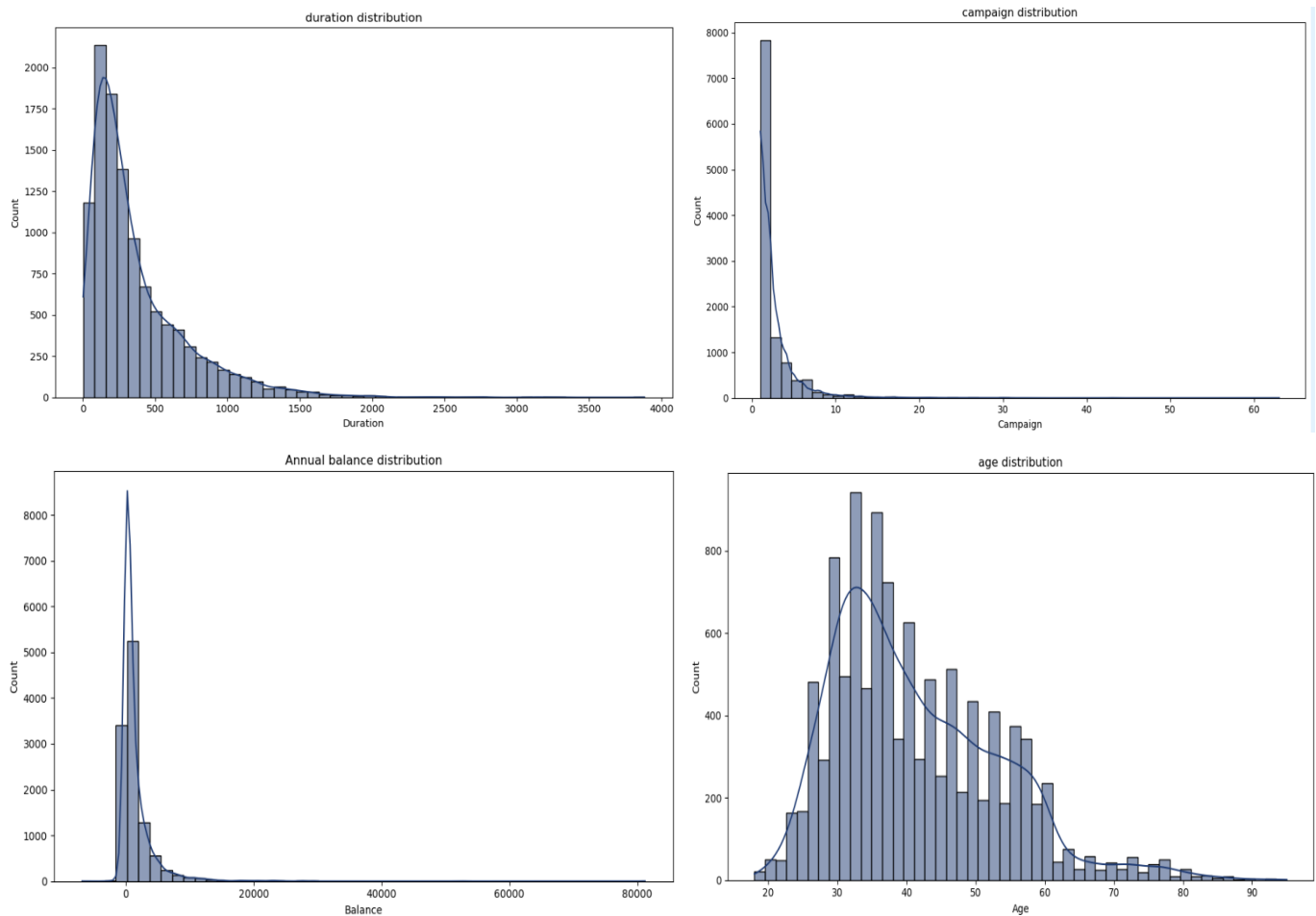


Figure 5.1 represents the distribution of subscriptions of the individuals who were participated to the data collection. If the subscription is ‘yes’ the customer had subscribed to the term deposits. Otherwise ‘no’. According to the analysis, 47.4% percentage of customers were subscribe to the term deposits and 52.6% were not.

Figure 5.2: Distributions of duration, campaign, balance and age



The figure describes Kernel Density Estimation plots of duration, campaign, balance and age. Duration distribution shows the distribution of call duration in seconds. Campaign distribution shows the number of times the bank contacts the customer for the campaign during the campaign period. Most of the customers were contacted lesser number of times however, some customers were contacted more than 60 times. The reason for that can be, the high potential customers because they are having high balance, stable job, and positive responses for previous campaigns or good customer profile. The annual balance distribution shows yearly average balance in euros. Most of the customers are in lower annual balances, but there are some customers who having higher annual balance. The reason for that can be that they are doing well paid jobs or having great management quality about their savings. The age distribution shows the ages of the customers who were involved to the bank campaign. According to this, most of customers are in 30 – 40 years. All of the plots are right skewed.

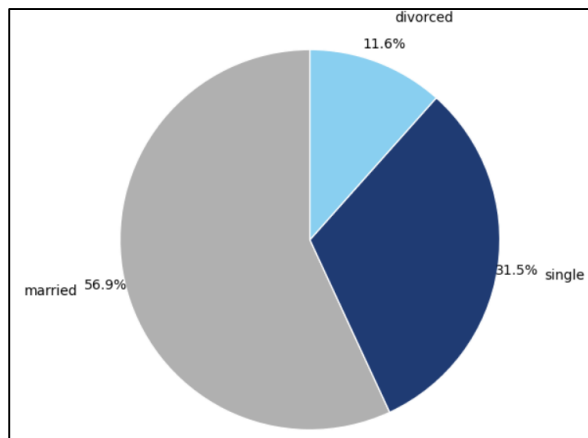
Figure 5.3: Distribution of marital status

Figure 5.3 explains the marital status of the customers. According to the plot, the percentage of 56.9% customers are married, 31.5% percent customers are single, and 11.6% customers are divorced.

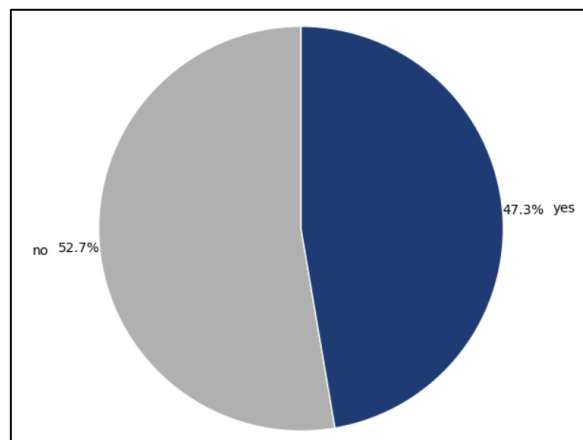
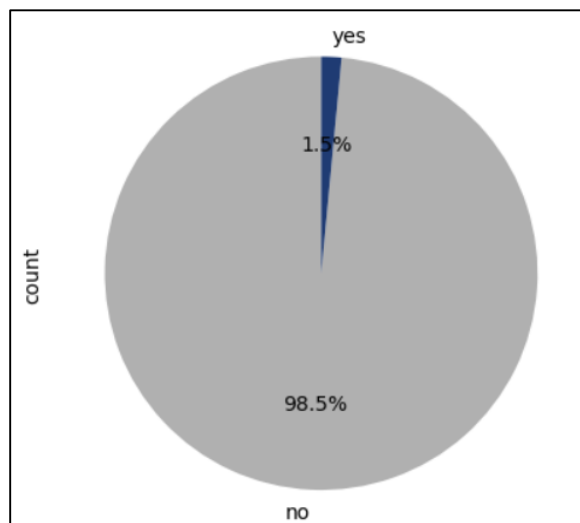
Figure 5.4: Distribution of housing loans

Figure 5.4 explains the customers who have a housing loan. In this dataset, 47.3% percent of customers have a loan for housing.

Figure 5.5: Distribution of default credits

The figure 5.5 describes the customers who have credits in default. According to this, 98.5% percent of customers do not have credits in default.

figure 5.6: distribution of communication type

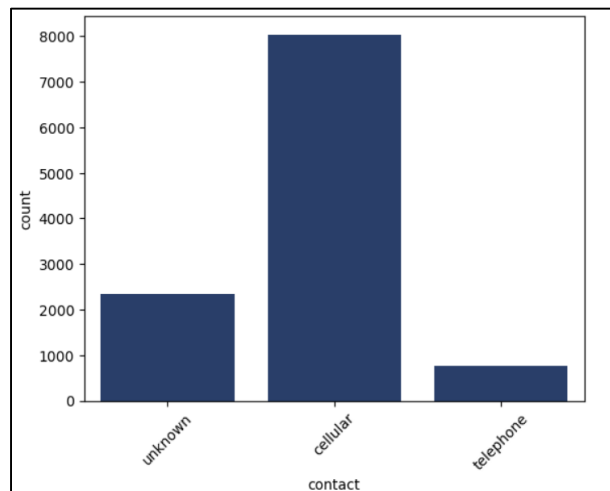


Figure 5.6 shows the type of communication customers used. The largest number of customers was contacted through cellular phones, and the smallest number of customers was contacted through telephones. The unknown category can be the customers whose communication details were not recorded by the bank, or their details were unavailable.

Figure 5.7: Distribution of job

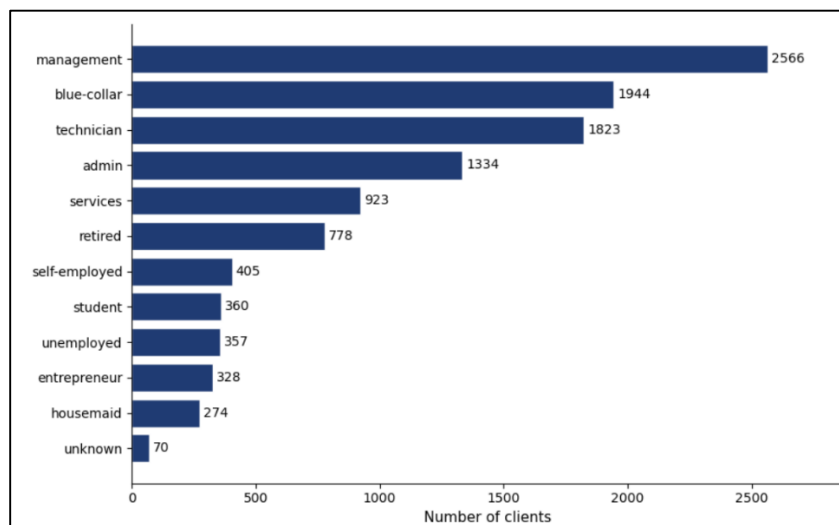


Figure 5.7 describes the job types customers have. The largest number of customers are doing management side jobs, and the least number of customers are housemaids.

Bivariate Analysis

Figure 5.8: deposit VS marital status

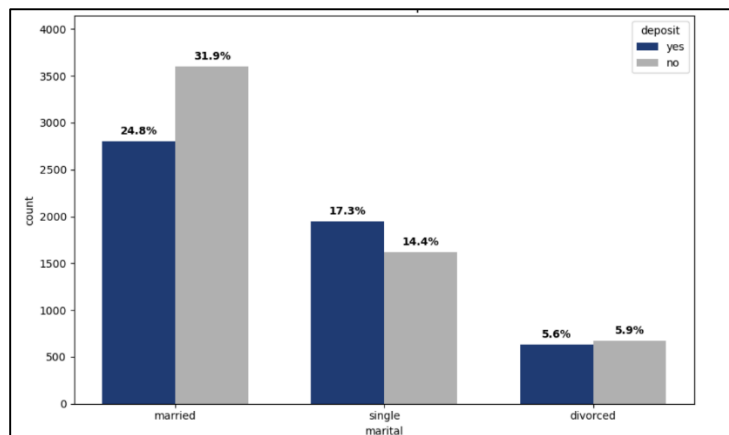


Figure 5.8 represents the subscription to term deposits across marital status. Most of the single customers tend to subscribe to the term deposit, and the majority of the married customers do not subscribe. Subscription of the divorced category is slightly equal to not subscribing.

Figure 5.9: deposit VS age

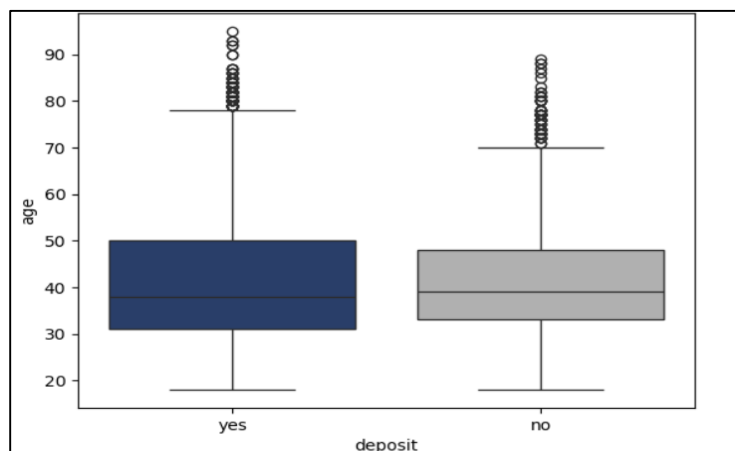


Figure 5.9 explains the age distribution across the term deposit subscription. Median of age of customers who subscribed is slightly equal to those who did not. The interquartile range of age of customers who subscribed is greater than that of those who did not.

Figure 5.10: deposit VS duration

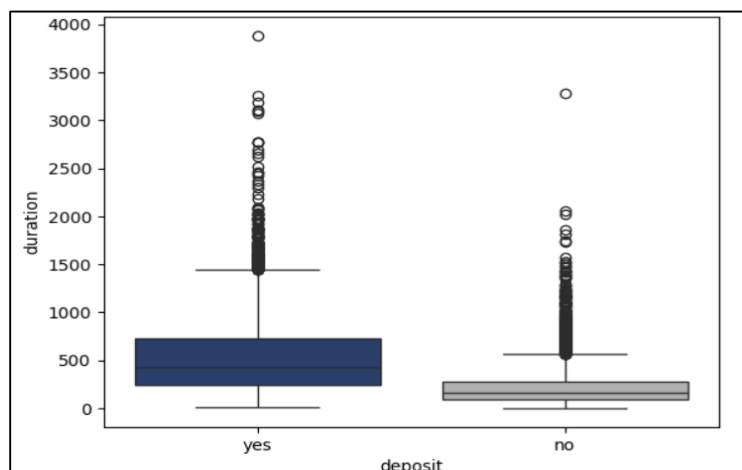


Figure 5.10 describes the call duration at the last contacted time at the campaign. Median of call duration is higher with the customers who subscribed. Median call duration of customers who didn't subscribe is close to zero. There are so many outliers in both groups. The reason for that may be different customer behavior.

Figure 5.11: deposit VS balance

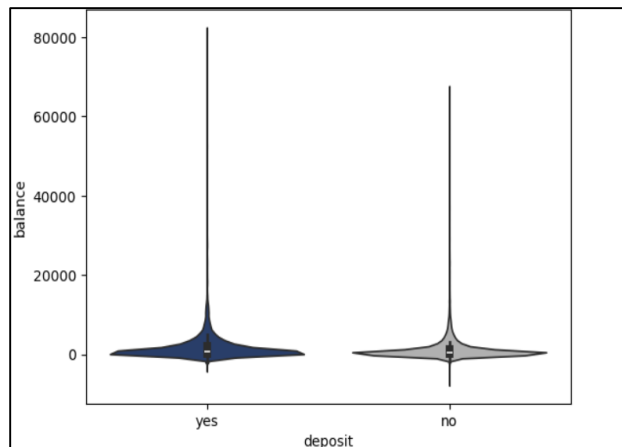


Figure 5.11 shows the variation of subscription across the annual account balance in euros. This is highly right skewed, indicating some have considerably larger balances. Customers who subscribed to the term deposit generally showed slightly higher account balances compared to non-subscribers. Therefore, financial capacity may positively influence to subscription.

Figure 5.12: deposit VS poutcome

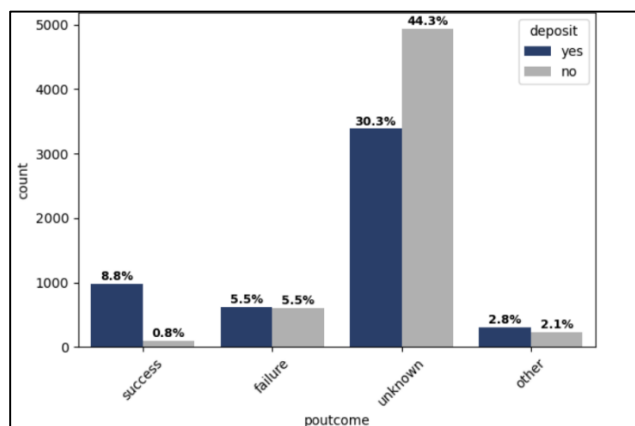


Figure 5.12 shows the impact of past campaign outcome on the current subscription. Most of the customers who gave positive responses for the past campaign, also subscribed to the term deposit. The customers whose past response were negative are equally contributed in to the subscription. From unknown category identifies newly contacted customers and other category represents customers whose

details are missing or unclear.

Figure 5.13: deposit VS contact

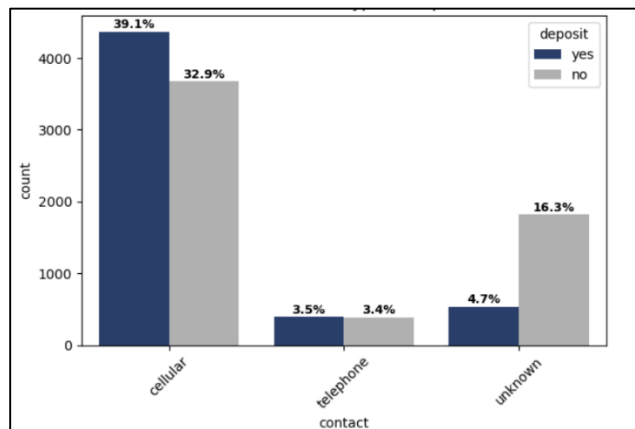


Figure 5.13 explains the impact of the communication method on the subscription to the term deposit. Most of the customers who received calls through their mobiles are more likely to subscribe. The least number of customers contacted through telephones, according to the recordings, the communication type of some customers is not recorded. The reason for cellular achieving great performance, may be when

contacting through telephone, sometimes a call does not reach the intended party.

figure 5.14: deposit VS housing

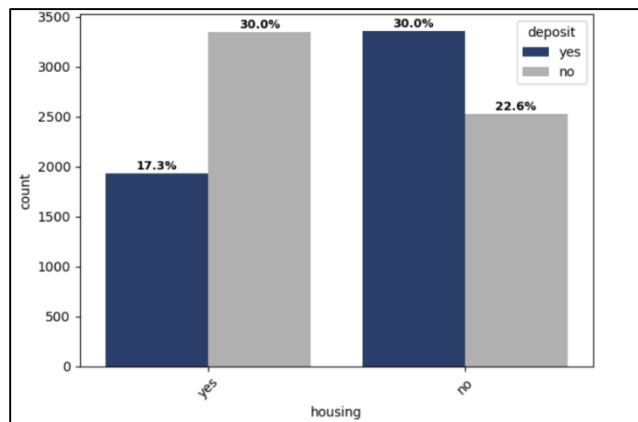


Figure 5.14 explains effect of having housing loan to the subscription. According to the plot, the customers who do not have housing loan tends to subscribe to term deposit. Among customers who are having housing loan, most of them did not subscribed.

In summary, the exploratory data analysis revealed several important patterns related to customer behavior. Single customers showed a higher tendency to subscribe to term deposits compared to married customers, whose subscription was relatively low. Call duration also showed a strong relationship with subscription outcomes, where longer conversations were associated with higher subscription outcomes. In addition, customers who subscribed generally maintained slightly higher account balances than others. Customers with positive outcomes from previous marketing campaigns were also more likely to subscribe again. Regarding communication methods, most customers were contacted through cellular communication, and this group showed a comparatively higher subscription rate, suggesting that contacting customers through cellular can achieve higher positive responses because, through cellular, it is easy to reach customers other than contacting through telephones. Furthermore, the customers without housing loans explained a greater tendency to subscribe to term deposits. Overall, these findings provide valuable business insights that can help banks improve customer targeting and marketing campaign effectiveness.

6. Advanced Analysis

Advanced analysis was conducted through four machine learning models. Through the model, predicting the term deposit subscription was aimed. As the study is a classification, Logistic Regression, Decision Tree model, Random Forest model and XGBoost model were built. Each model was trained and evaluated based on accuracy, recall, precision and F1-score. For the further improvement of the models, hyperparameter tuning, threshold changing and transformation techniques were used.

Cross validation and hyperparameter tuning were performed to improve model and reduce overfitting. GridSearchCV was applied for Logistic Regression and Decision tree models, while RandomizedSearchCV was used for Random Forest and XGBoost due to larger parameter spaces.

6.1 Model Performance

Figure 6.1: comparison between baseline and tuned models

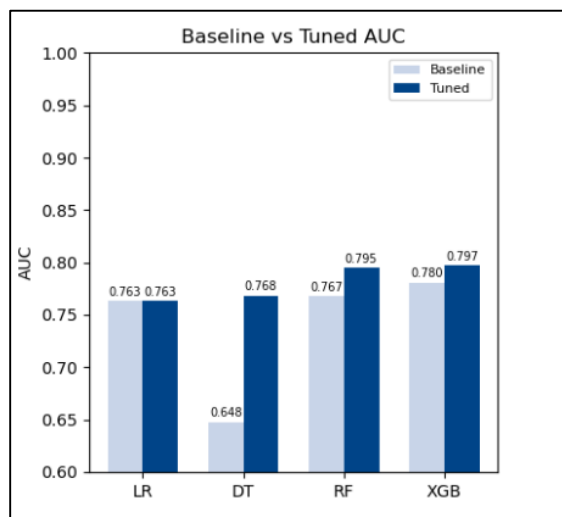
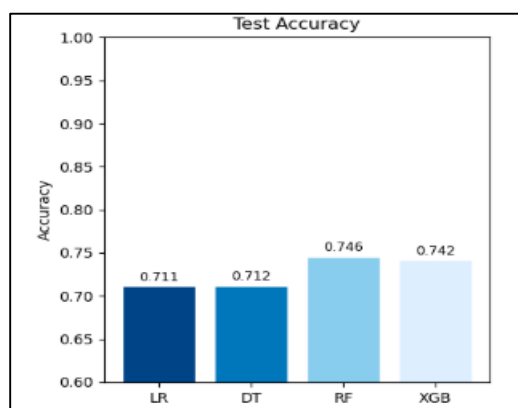


Figure 6.1 compares the baseline and tuned ROC AUC scores of the evaluated models. Hyperparameter tuning improved the performance of most models. The Decision Tree showed the largest increase in AUC from 0.648 to 0.768. Random Forest also achieved noticeable improvement. Logistic Regression showed minimal change after tuning, indicating that baseline model was already performing near its optimal performance.

Figure 6.2: Test accuracy of the tuned models



The figure 6.2 represents the variation of the accuracy of four machine learning models. Logistic Regression achieved an accuracy of 71.1% and Decision Tree achieved 71.2%. Random Forest (RF) produced the highest accuracy of 74.6%, indicating that the best overall classification performance among the evaluated models. XGBoost (XGB) achieved an accuracy of 74.2%, performing slightly below than random forest but it is still outperforming. The results indicate that ensemble learning methods particularly Random

Forest and XGBoost, provided superior predictive performance compared to Logistic Regression and Decision Tree models.

6.2 Model Comparison

Table 6.2.1: Model Comparison

Model	Accuracy	Recall	Precision	F1 score	ROC AUC
Logistic Regression	0.71	0.64	0.72	0.68	0.76
Decision Tree	0.71	0.55	0.78	0.65	0.76
Random Forest	0.75	0.60	0.82	0.69	0.80
XGBoost	0.74	0.63	0.79	0.70	0.80

The results indicate that Random Forest and XGBoost achieved the strongest predictive performance. Both models obtained the high accuracy and the highest ROC-AUC scores, demonstrating better capability in distinguishing between customers who subscribed to deposits and those who did not.

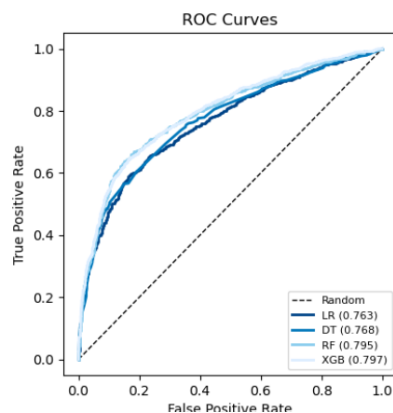
Random Forest achieved the highest accuracy (0.75) and precision score (0.82), indicating strong reliability when predicting deposit subscribers. XGBoost achieved the highest F1 score (0.70), reflecting a balanced tradeoff between precision and recall.

Logistic Regression produced stable and balanced results across all evaluation metrics while maintaining high interpretability. This makes it is valuable for understanding the influence of customer characteristics on deposit subscription behavior and gain business insights.

Decision Tree achieved performance comparable to Logistic Regression in terms of accuracy and ROC-AUC, but its lower recall score suggests a weaker ability to identify potential deposit subscribers.

Overall, based on ROC AUC, models are adequate. Random Forest model provides the highest predictive accuracy, whereas Logistic Regression offered greater explanatory value for business decision making and marketing strategy development.

Figure 6.2.1: ROC Curves



The ROC curve shows how well each model separates into two classes. According to the plot, all models are well above 0.5. Therefore, all four models are not random.

6.3 Feature Importance

Although the Random Forest model is the best predictive model for this study, its accuracy and ROC AUC values are slightly greater than the Logistic Regression model. For discussing feature importance and business insights, the most suitable model is the Logistic Regression model because it shows the direction of the relationship, the strength of influence and the positive or negative impact.

Figure 6.3.1: Feature Importance by Logistic Regression model.

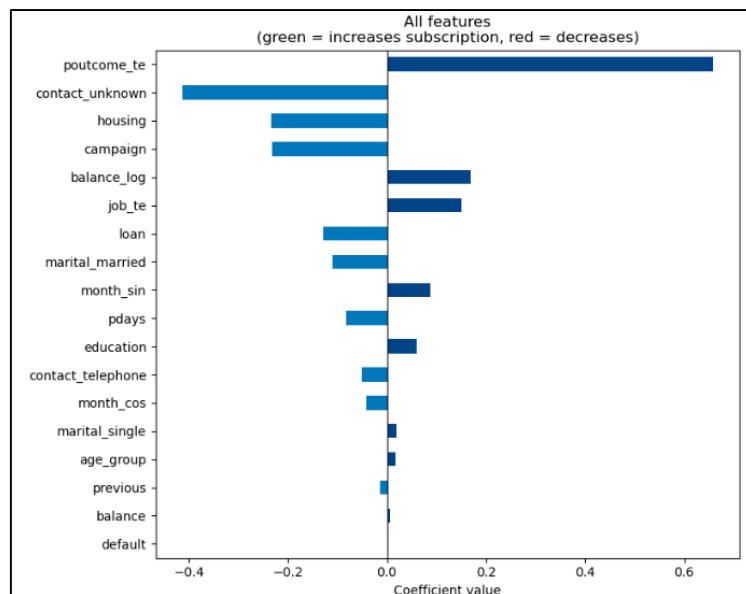
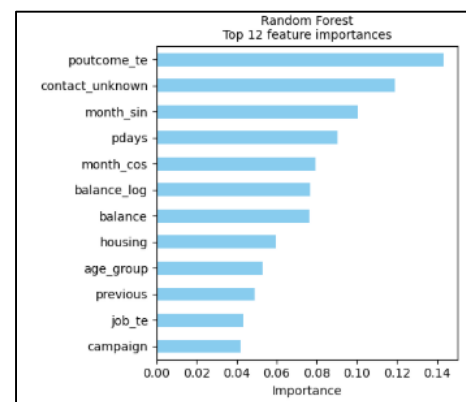


Figure 6.3.2: Feature Importance by Random Forest model



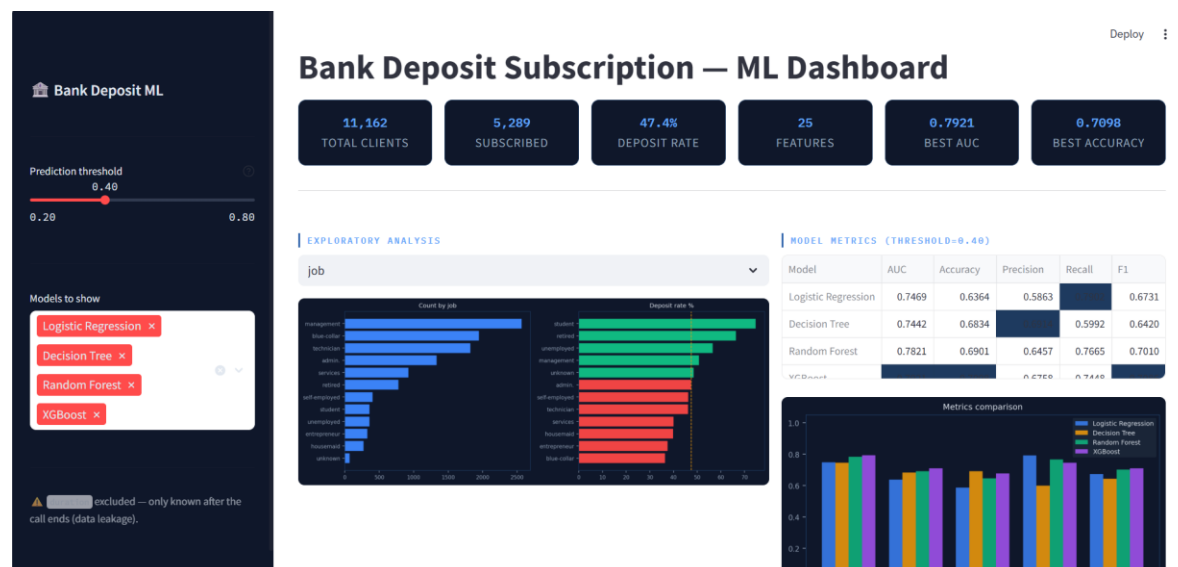
The feature importance chart highlights which variables contributed important 6most to predict subscription to term deposits. The most influential feature was poutcome suggesting that the customers who gave positive reactions for past campaign have higher preference to subscribe to term deposits. Then the contact variable has negative impact on the subscription to term deposits. In the explanatory data analysis, the *figure 5.13* shows that customers whose communication type cannot be verified tend not to accept the term deposit. Housing, campaign and month variables have moderate impact on the subscription. *Figure 5.14* clearly shows that the most of the customers who do not have housing loans subscribed the deposit. Other moderately important features including balance, job, and loan also indicate that financial status play a role on decision of subscription. Marital status also has a moderate impact on subscription. *Figure 5.8* shows the married customers did not subscribe and most of the single customers subscribed the term deposit.

Overall, the feature importance analysis revealed that previous campaign outcomes, customer financial characteristics, and contact related variables were the most influential factors affecting deposit subscription behavior. Logistic regression provided clearer interpretability for business insight generation, while Random Forest identified the most predictive variables contributing to classification performance.

6.4 Dashboard Creation

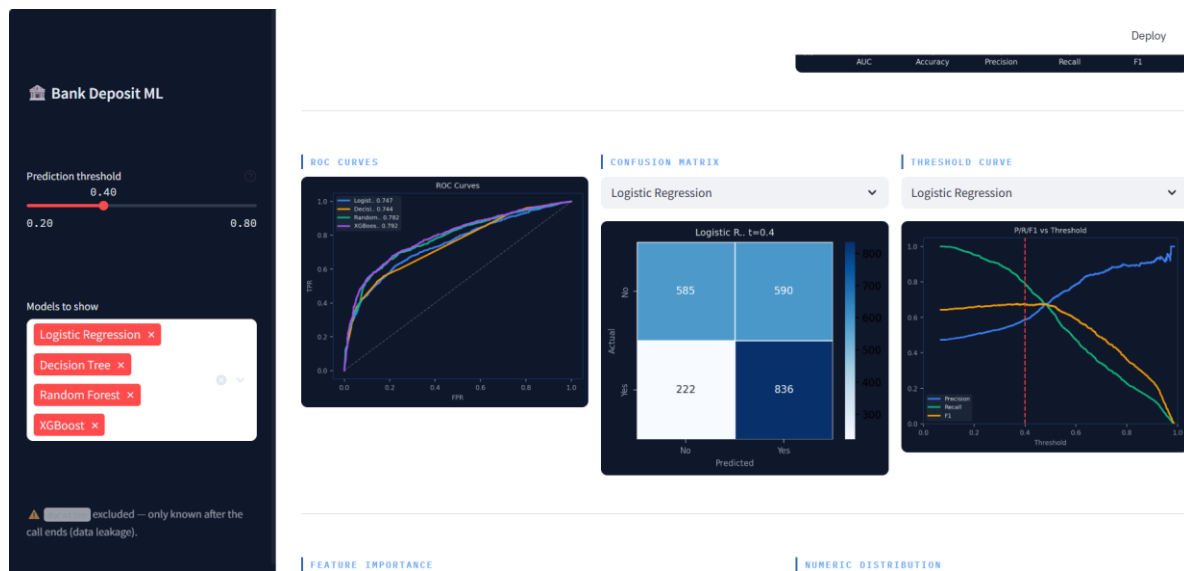
The dashboard was created to improve practical usability and to visualize efficiently. The dashboard board illustrates customer behavior patterns, model performance, and key marketing insights of the subscription to a term deposit. Also, through this dashboard, bankers can predict the customer behavior towards the subscription.

Figure 6.4.1: dashboard interface (part 1)



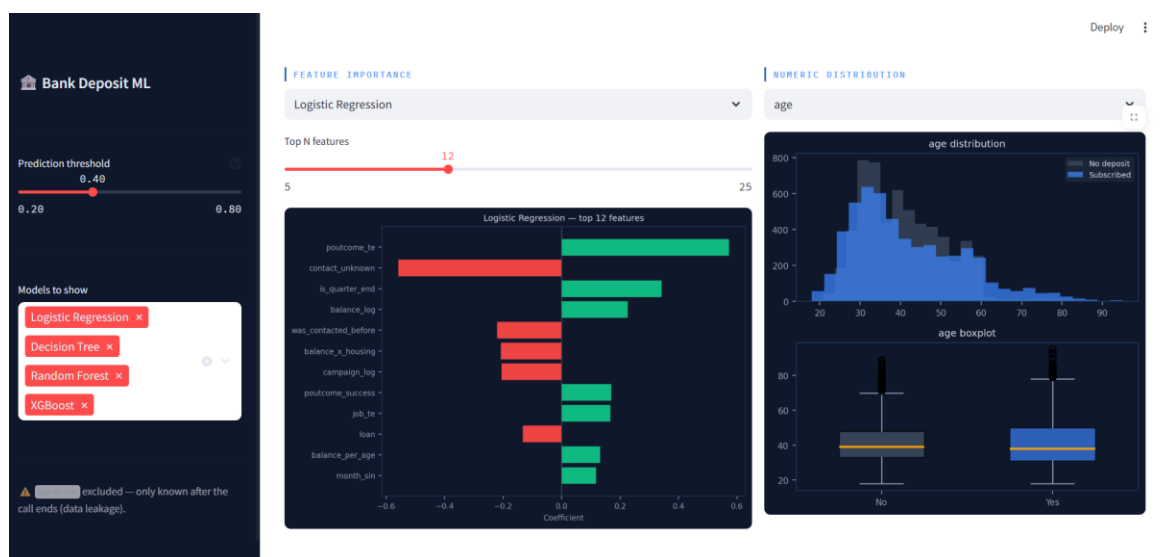
The metrics provide basic idea about the total clients, deposit rate and the model performance. Then the explanatory data analysis was conducted. Selecting different features, can obtain the distribution of the particular feature and the impact of the feature on the subscription to term deposit.

Figure 6.4.2: dashboard interface (part 2)



Then the model performance was analyzed. From the confusion matrix and threshold curve, can obtain individual performance of each model by choosing the model.

Figure 6.4.3: dashboard interface (part 3)



From this feature importance can be analyzed. Also can obtain distributions of the numerical features. These two plots also interactive visualization plots.

Figure 6.4.4: dashboard interface (part 4)

The screenshot displays a web interface for a machine learning model titled "Bank Deposit ML". On the left, a sidebar contains a "Prediction threshold" slider set to 0.40, a list of models (Logistic Regression, Decision Tree, Random Forest, XGBoost), and a warning about data leakage. The main area is titled "PREDICT FOR A NEW CLIENT" and features a grid of input fields for various customer attributes. A "Deploy" button is in the top right. At the bottom, a red "Predict" button is visible.

Age	Education	Personal loan	Contacts this campaign
40	secondary	yes	2

Job	Balance (€)	Credit default	Days since last contact (-1=never)
admin.	1000	no	-1

Marital	Housing loan	Contact type	Previous contacts
married	yes	unknown	0

Previous outcome	Model
unknown	Logistic Regression

Predict

At last, prediction option was included. From that, banker can get idea a particular customer will accept the term deposit or not. By entering the information model ask, can obtain the prediction of the customer. In here also have option to change the model as preferred.

7. General Discussion and Conclusion

7.1 General Discussion

- The analysis showed that customer financial characteristics and previous marketing outcomes had a strong influence on deposit subscription behavior.
- Exploratory data analysis revealed noticeable differences in subscription patterns across variables such as outcome, housing, contact, marital, and age.
- Feature importance analysis showed that previous campaign success, customer balance, contact information, and campaign-related variables were among the most influential predictors.
- Although the duration variable had a strong impact on deposit subscription prediction, it was excluded from final models to avoid data leakage and improve practical real world applicability.
- Random Forest model is the most appropriate model due to the highest predictive accuracy (0.75) and strong discrimination ability among the evaluated models.
- The performance of XGBoost and Random Forest was very similar. Random Forest was selected as the preferred model because it achieved comparable predictive performance with lower model complexity and easier implementation.
- Logistic Regression demonstrates stable and balanced performance while providing greater interpretability for business insight generation.
- Compared to other studies, this study mainly focused on improving a realistic, interpretable model rather than only focusing on the accuracy of the model.

7.2 Conclusion

- The study successfully identified important factors influencing customer deposit subscription behavior. Such as previous campaign outcomes and financial stability.
- Random Forest achieved highest classification accuracy and Logistic Regression provided valuable interpretability and supported the making meaningful business insights.
- Through the dashboard the banker can find the potential customers easily.
- The findings highlight the potential of predictive analytics to improve customer targeting and marketing efficiency in banking campaigns.
- This study was conducted based on one region, which may limit the generalizability of the findings to different banking environments. In addition, although the removal of the duration variable improved the practical applicability of the models, it may have reduced the overall predictive performance.
- Future studies can incorporate larger and more diverse datasets, additional variables to further improve predictive performance.

8. REFERENCES

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